High Severity Alert Investigation

Findings

- **Alert:** Compromised account conducting hands-on-keyboard attack. Severity High, status Detected, tagged Attack Disruption.
- **Time:** first and last activity 2026-10-05 4:06:41 PM (23:06:41 UTC). Alert generated at 4:09:01 PM (23:09:01 UTC).
- **Host:** MDF-DT-W11 (Windows 11). **Account:** AzureAD\UserA.
- **Detection:** source EDR, service Microsoft Defender for Endpoint, detection technology Behavior and Machine Learning.
- **ATT&CK:** T1053.005 Scheduled Task. Categories: Execution, Privilege Escalation, Persistence.
- **Triggering event:** `schtasks.exe` (PID 7964, High integrity, full token elevation) created a scheduled task with: `SCHTASKS /Create /SC ONCE /TN spawn /TR C:\windows\system32\cmd.exe /ST 20:10`
- **Artifacts:** task name `spawn`; task action `C:\windows\system32\cmd.exe`; trigger once at 20:10; `schtasks.exe` at `C:\windows\system32\schtasks.exe`, SHA1 `dee1b9222da8463ea02cfd76efd4371fea3f88e3`.
- **Automated response:** Contain user on UserA at 4:09 PM, submitted by Attack disruption, status Completed.
- **Incident:** the alert belongs to "Hands-on keyboard attack was launched from a compromised account (attack disruption)". **TODO:** incident ID and alert count from the incident page.
- **Verdict: true positive for the behavior, benign in intent.** The activity is almost certainly the authorized Atomic Red Team test T1053.005 (test 2, "Scheduled task Local") that I ran from an elevated PowerShell session. The alert's "compromised account" framing is not supported by the evidence: I found no sign of a third party.

Investigation Summary

At 4:06:41 PM (23:06:41 UTC), `schtasks.exe` created a run-once scheduled task named `spawn`, set to launch `cmd.exe` at 8:10 PM (20:10) on the VM's local clock, about four hours later. Defender XDR raised a High severity alert at 4:09:01 PM (23:09:01 UTC), and automatic attack disruption ran Contain user on UserA at 4:09 PM (23:09 UTC).

Why it fired. The alert's story contains one event, the task creation. The same incident also holds the two "Anomaly detected in ASEP registry" alerts from 4:01 PM, so the earlier persistence activity likely raised Defender's confidence, but I have not verified how it weighted

them. Microsoft describes automatic attack disruption as combining signals from several sources to decide an asset is compromised, then containing it automatically.

Is it expected? Yes. My Atomic Red Team output shows test T1053.005-2 completing with "SUCCESS: The scheduled task "spawn" has successfully been created" and exit code 0. I started it shortly after 11:03:52 PM UTC, and the alert event is at 11:06:41 PM UTC. The task name, run-once schedule, and `cmd.exe` action are consistent with that test's definition.

What happened before and after (shift left).

- 3:49 PM (22:49 UTC): alerts from the Atomic Red Team install (PowerSploit and Meterpreter post-exploitation tool, and two Suspicious behavior by cmd.exe), most likely caused by the downloaded test content.
- 4:01:55 to 4:01:59 PM (23:01:55 UTC to 23:01:59 UTC): Run and RunOnce registry writes from the T1547.001 tests (T1547.001, T1112).
- 4:06:41 PM (23:06:41 UTC): the scheduled task creation. At 4:06:41.919 to 4:06:42.777 PM (23:06:41.919 to 23:06:42.777 UTC), `powershell.exe` and `cmd.exe` events tagged T1053 Scheduled Task/Job follow.

The sequence is what a real human-operated intrusion could look like, which is why Defender escalated it.

Scope. Entra lists one device in the tenant, and UserA's only associated interactive logon device in the last 30 days is MDF-DT-W11. There are no other hosts to search.

5W1H

- **Who:** AzureAD\UserA, the lab's non-admin user, working in an elevated PowerShell session.
- **What:** a scheduled task `spawn` was created with `schtasks.exe` to run `cmd.exe` once at 20:10 (T1053.005). Defender flagged it High and contained the account.
- **When:** October 5, 2026, 4:06:41 PM (23:06:41 UTC). Not ongoing: the test was removed with `-Cleanup` the same afternoon and the containment was undone.
- **Where:** MDF-DT-W11, the lab Windows 11 endpoint and the only Entra device in the tenant.
- **Why:** detection validation for the Day 20 assignment. There is no evidence of malicious intent.
- **How:** `Invoke-AtomicTest T1053.005 -TestNumbers 2` after installing Invoke-AtomicRedTeam.

Recommendations

If this were a real alert instead of a test:

1. **Treat it as a true positive until proven otherwise.** Ask the user whether they created the task, and determine whether it ran: look for a `cmd.exe` launched by the Task Scheduler service around 20:10.
2. **Review UserA's sign-ins before restoring the account.** Check `EntraIdSignInEvents` for anything off-baseline.
3. **Remove the task and sweep other persistence.** Here the same incident also showed Run/RunOnce persistence, so one persistence finding should prompt a search for others.
4. **Scope across the environment.** Search for `schtasks.exe /Create` with `/TR cmd.exe`, and for the task name, on every onboarded device.

Supporting evidence

Day20-05-t1053-005-run.png
The test creating `spawn`, with exit code 0

```
PS C:\WINDOWS\system32>    (Get-Date).ToUniversalTime()

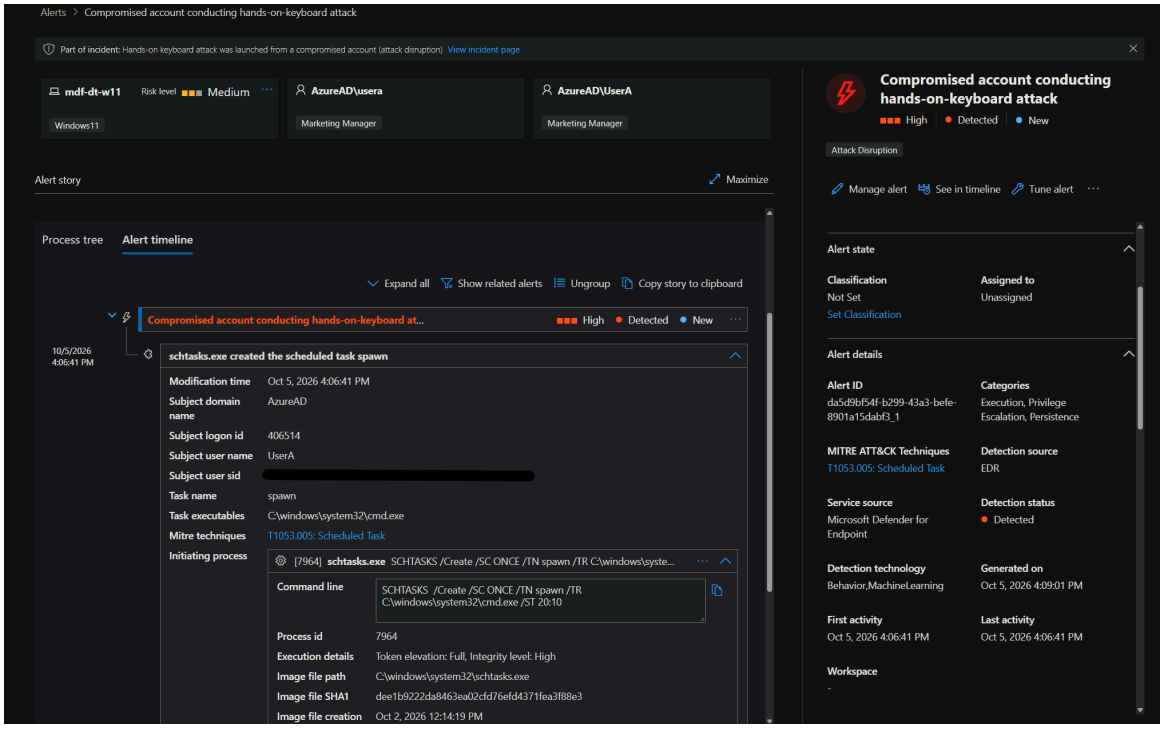
Monday, October 5, 2026 11:03:52 PM


PS C:\WINDOWS\system32> Invoke-AtomicTest T1053.005 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

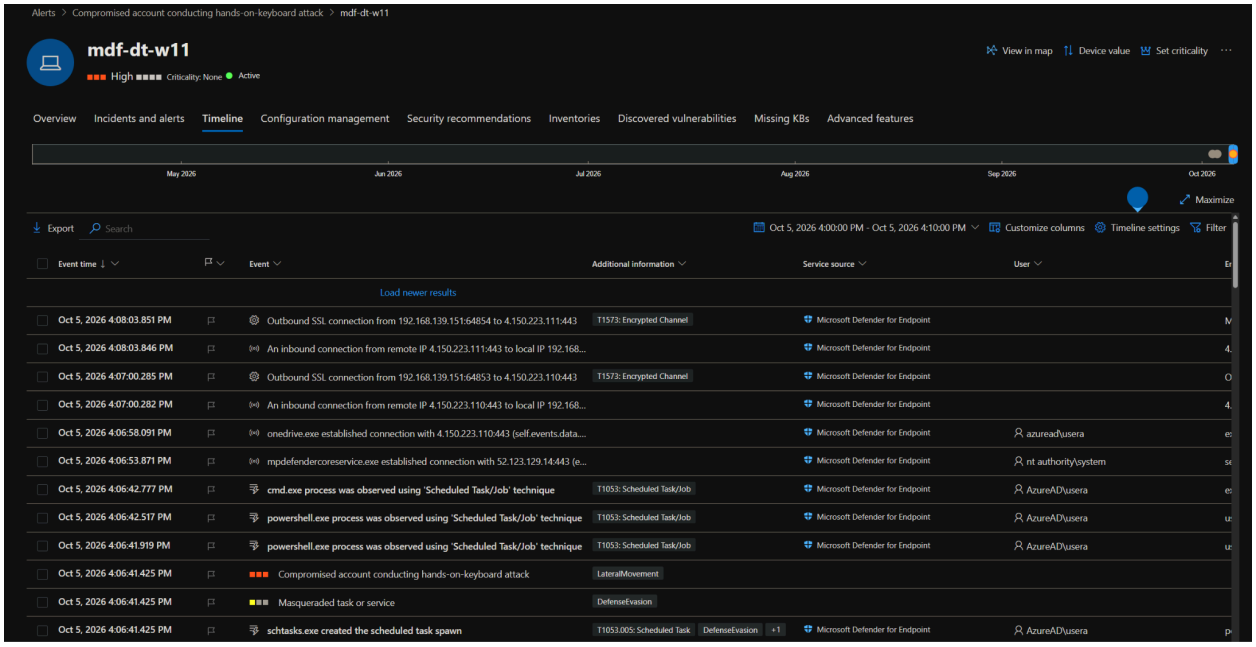
T1053.005-1 Scheduled Task Startup Script
T1053.005-2 Scheduled task Local
T1053.005-3 Scheduled task Remote
T1053.005-4 Powershell Cmdlet Scheduled Task
T1053.005-5 Task Scheduler via VBA
T1053.005-6 WMI Invoke-CimMethod Scheduled Task
T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
T1053.005-8 Import XML Schedule Task with Hidden Attribute
T1053.005-9 PowerShell Modify A Scheduled Task
T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
T1053.005-11 Scheduled Task Persistence via CompMgmt.msc
T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
T1053.005-13 Turla Topinambour Dropper and Scheduled Task Persistence
T1053.005-14 Turla KopiLuwak Scheduled Task for JavaScript Stager
PS C:\WINDOWS\system32>    Invoke-AtomicTest T1053.005 -TestNumbers 2
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-2 Scheduled task Local
SUCCESS: The scheduled task "spawn" has successfully been created.
Exit code: 0
Done executing test: T1053.005-2 Scheduled task Local
PS C:\WINDOWS\system32>
```

Day20-07-alert-detail.png
The High alert story: `schtasks.exe` command line, task details, ATT&CK mapping



Day20-08-device-timeline.png
Timeline 4:00 to 4:10 PM (23:00 to 23:10 UTC) with ATT&CK tags



Day20-06a-action-center-contain-user.png
Contain user submitted by Attack disruption

